# CS 4650/7650 ECE 4655/7655 Digital Image Processing
# Fall 2024
# Homework 1B: Point Processes [60 pts]

**Out:** Thursday Sep 5

**Due:** Tuesday Sep 17 (Midnight)

**The goals of this first assignment are**:
(1) trying out basic image read, write, access, display and plotting functions; and
(2) review of point process (intensity transformation) concepts.

**Programming Language**: Python (recommended) or C/C++

**[30 pts] Part B1 Histogram computation**

Write a function to compute image histogram of a graylevel image. This function should be able to generate histograms for different number of bins not just nbins=256. Try to minimize computation.

function [H]=myimhist(I,nbins)

Note: You are not allowed to use built in functions for histogram computation.

a) Modify the function above so that it computes histogram of a region specified using a binary mask (compute histogram of the image pixels where mask is equal to 1).

function [H]=myimhist(I,nbins,mask)

For (a)&(b) display

- the input images,
- input masks,
- plot the obtained histograms (Hint: you can use stem/bar/plot functions etc.),
- report timing (you can use time, timeit functions).

**[30 pts] Part B2 Contrast enhancement**

a) Minimum-Maximum Linear Contrast Stretch: Read a grayscale image. Determine the intensity limits of the image. Apply linear stretch to the image. Show original and enhanced images and plot corresponding histograms.

b) Now re-compute the upper and lower intensity limits of the image but this time discard some percent of the highest and lowest valued pixels. Re-apply linear contrast stretch. First

discard 1% then 5% of the image pixels from both ends of the intensity spectrum and apply linear contrast stretch.
Show input/output images and plot input/output histograms. Comment on the results. **Hint:** Use histogram or cumulative histogram to determine the new limits.

**Note:** Discard amounts (i.e. 1% or 5%) will be calculated in terms of the total number of pixels in the image, not 1% or 5% of the intensity range.

For this question you can use built-in histogram functions, but do not use built-in histogram adjustment functions.

**Report Structure:** Your Written Report must conform to the following format.

1. Header
    - Course number and name
    - Assignment number and title
    - Your name
    - Date
2. Abstract (2 to 4 sentences describing the problem and the major results obtained)
3. Introduction (A brief statement of the experiments to be performed and the goal of the assignment)
4. Experiments and Results (figures, tables, graphs, …). You need to use given test images. See the HW1B_TestTemplate for specific test parameters and structure of the report.
5. Discussion/Conclusions
6. References/Appendix (As needed)

**References**


1. General tools that are useful to look at your input and output images: NIH ImageJ or Fiji which is an extension of ImageJ (opensource Java), imageMagick (opensource), gimp (opensource C++), Paint.Net, irfanview (shareware), Adobe Photoshop, Preview (MacOS), Windows PhotoViewer.
2. Lect3_IntensityTransformations.pdf
3. https://numpy.org/doc/stable/user/numpy-for-matlab-users.html